

Multi-layer security system for Large Language Models:

Target:

- **Reduce prompt injecting**
- **Reduce Hallucination**
- **Preventing leakage of private data**
- **Reduce Biased Responses**

Work Flow →

User prompt → Injection detection → Prompt cleaning → Kernel processing → VectorDB search → RAG → Base LLM response → Cross-LLM evaluation → Publish or block

1. Transformer classifier for Injection Detection
2. Kernel tries to understand the meaning
3. Converts into chunks and vector comparison with stored vector and RAG then combines the output
4. Cross LLM verification after base LLM

Do not trust the user prompt, do not trust the model's internal knowledge, and do not trust the first generated answer.

2. Prompt Injection Literature Review

- **Prompt injection**
- **Jailbreaking : Particular prompt injection to bypassing safety**

Most papers mainly study:

- **Instruction Hijack**
- **Policy Bypass**

Much less research focuses on:

- Data stealing
- Tool misuse
- Long-term memory poisoning

Workflow →

Prompt Filtering → RAG → Fine tuning the model response and tone

3. Multi-Layered Framework for LLM Hallucination Mitigation

Target

- Reduce LLM hallucination
- Increase factual accuracy
- Make LLM safer for high-risk applications
- Know when the LLM should **not answer**

Work Flow →

Prompt Engineering → RAG → Fine-Tuning → Confidence Check → Answer or Escalate

4. Scheduling DL and LLM Training Jobs in GPU Clusters

If GPU resources are poorly managed:

- GPUs stay idle
- Training takes longer
- Cost increases
- Some jobs wait too long
- Users may get unfair access to resources

Main Target

- Better GPU utilization
- Reduce training time
- Reduce Job Completion Time (JCT)
- Reduce GPU waste
- Support multiple training jobs
- Improve fairness
- Reduce training cost

All about making training efficient using gpu properly.

*****Mitigating Insider Threat: A Neural Network Approach for Enhanced Security**

This paper tries to detect insider threats inside IoT-enabled organizations.

The model tries to:

- Detect malicious insiders
- Reduce false alarms
- Solve data imbalance
- Reduce data dimensions
- Improve data quality
- Improve DNN performance

User Activity Logs → Preprocessing → IPCA(Improved Principal Component Analysis) → K-Means Clustering → EBiGAN → DNN-PI → Malicious / Non-Malicious

*****Protecting Privacy in Knowledge Graphs With Personalized Anonymization**

This paper is about protecting users' privacy when a Knowledge Graph is shared or published.

Different user may want different type or level of privacy.

Main Target

- Protect user identity
- Allow each user to choose their own privacy level
- Prevent re-identification

- Keep as much useful Knowledge Graph information as possible
- Reduce information loss

WorkFlow

Knowledge Graph → User k Values → Distance Calculation → VAC(Varying-Anonymity Clustering Algorithm: Groups similar user) → Merge-Split → KGG(Generalizes relationship in/out degrees) → Anonymized KG